Analog Multi Wave Signal Generator

ECE 145 - Final Draft Proposal, Professor Schmitz

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Abstract:

Our goal is to design and build a hardware function generator that produces square, triangle, and sine waves without a microcontroller in the signal path. The instrument will run from a single 12 V supply and will provide an adjustable output frequency of roughly 270 Hz to 12 kHz, with an independent amplitude control for each output. The square and triangle waves come from a relaxation oscillator built around the LM13700 dual operational transconductance amplifier (OTA), in which the oscillation frequency is set by a bias current that a front panel potentiometer controls. The sine wave is derived from the triangle wave with a differential pair shaping circuit that exploits the hyperbolic tangent transfer function of a bipolar transistor pair, followed by an op-amp low-pass stage. Three inverting op-amp output stages, all referenced to a buffered 6 V mid rail, provide amplitude adjustment that does not interact with the frequency setting. We will simulate each block, build and test the blocks individually on a breadboard, and then combine them into a compact instrument that can supply test signals for basic electronics experiments.

Background and Research

Our second draft proposal described the generator as a chain of an op-amp RC square wave oscillator, an integrator, and a sine shaper. The feedback we received asked for more detail on how each block would be implemented, in particular the frequency adjustment and the triangle to sine conversion. While researching those two questions we found that a single LM13700 dual OTA can produce both the square and the triangle wave with the frequency controlled by a current rather than by a variable RC network. This proposal is built around that approach. The sources below shaped the design.

1. Square and triangle wave generation (LM13700 datasheet, Texas Instruments). The datasheet's application section includes a triangle/square wave generator that uses one OTA as a current source integrator and the second OTA as a Schmitt trigger. The oscillation frequency is $f = I_C / (4\ C\ I_A\ R_A)$, where I_C and I_A are the two amplifier bias currents, C is the timing capacitor, and R_A sets the comparator threshold. Because I_C enters the formula linearly, a single potentiometer that sets I_C gives a wide, smooth frequency sweep.
   https://www.ti.com/lit/ds/symlink/lm13700.pdf
2. Triangle to sine conversion (Analog Devices, StudentZone, September 2021). Mercer and Miclaus describe an ADALM2000 activity that turns a triangle wave into an approximate sine wave by driving an NPN differential pair. The pair's transconductance follows a tanh function of the input voltage, which rounds the peaks of the triangle, and an op-amp differential stage with a small feedback

capacitor removes the remaining high-frequency content. A trim potentiometer in the emitter network is adjusted for the best symmetry and lowest even order distortion.
https://www.analog.com/en/resources/analog-dialogue/studentzone/studentzone-september-2021.html

3. Complete function generator architecture (Analog Devices application note AN-278). This note shows a full generator with separate waveform generation, buffering, and sine shaping stages. We used it to decide how the blocks should be partitioned and where buffers are needed to prevent one stage from loading another.
https://www.analog.com/media/en/technical-documentation/application-notes/444186898AN278.pdf
4. Op-amp multivibrator and integrator (Electronics Tutorials). These pages describe the RC astable multivibrator and the op-amp integrator that our second draft proposed. We keep them as a fallback: if the OTA oscillator proves difficult to bias, an op-amp astable followed by an integrator can produce the same two waveforms, at the cost of a frequency control that depends on a variable resistor that gives a smaller tuning range.
https://www.electronics-tutorials.ws/opamp/op-amp-multivibrator.html and https://www.electronics-tutorials.ws/opamp/opamp_6.html
5. Output stage op-amp (TL074 datasheet, Texas Instruments). The TL074 is a quad JFET-input op-amp with a 3 MHz gain bandwidth product and a 13 V/µs slew rate, which is sufficient for several volts of output swing at the top of our frequency range. Its output has built in short circuit protection. Two packages are used: one for the V_{ref} buffer and the shaper op-amp, and one for the three output amplifiers.
https://www.ti.com/lit/ds/symlink/tl074.pdf

The reference circuits in sources 1 and 2 are drawn for dual supplies (±15 V and ±5 V). A dual supply is inconvenient for a portable instrument, so our design runs from a single 12 V rail and replaces the reference ground of those circuits with a buffered 6 V mid rail voltage, V_{ref}. All component values in this proposal are calculated for the 12 V rail.

Design Overview:

Figure 1 shows the system divided into input, processing, output and power blocks. The user inputs are a frequency potentiometer, and one amplitude potentiometer per output. The processing section contains the OTA oscillator, which generates the square and triangle waves, and the differential pair shaper, which converts the triangle wave to a sine wave. The output

section consists of three inverting op-amp stages. A power block generates V_{ref} for every stage.

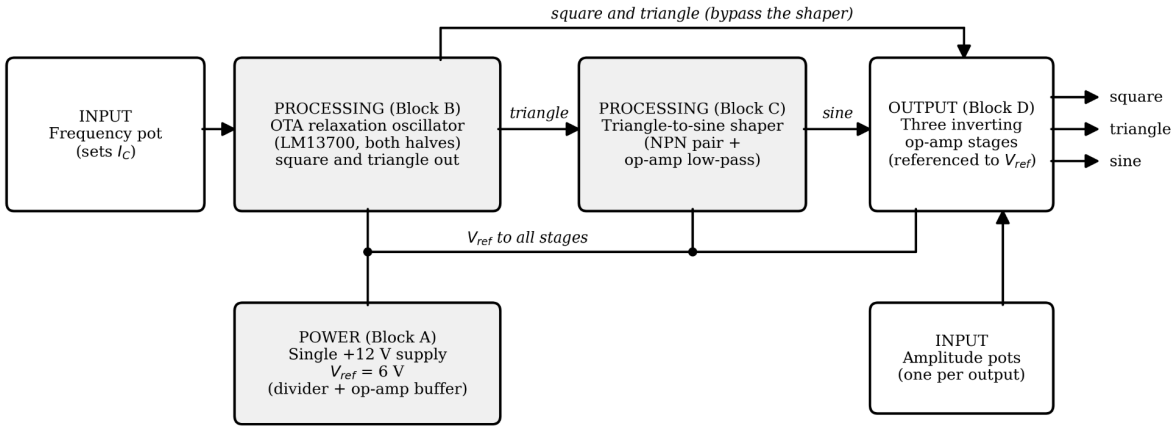


Figure 1. Block diagram of the analog function generator.

Block A: Power and 6 V reference

The instrument takes a regulated 12 V input from the bench supply during development and from a 12 V DC adapter in the final build. Bulk (47 µF) and ceramic (100 nF) capacitors decouple the rail. Two 10 kΩ resistors divide the rail to 6 V, a 10 µF and a 100 nF capacitor filter the divider, and one op-amp of the TL074 buffers the divided voltage so that V_{ref} can source and sink the currents drawn by the oscillator (up to ±0.2 mA through R_A), the shaper, and non-inverting inputs of the output amplifiers without drifting. This block is shown in Figure 2.

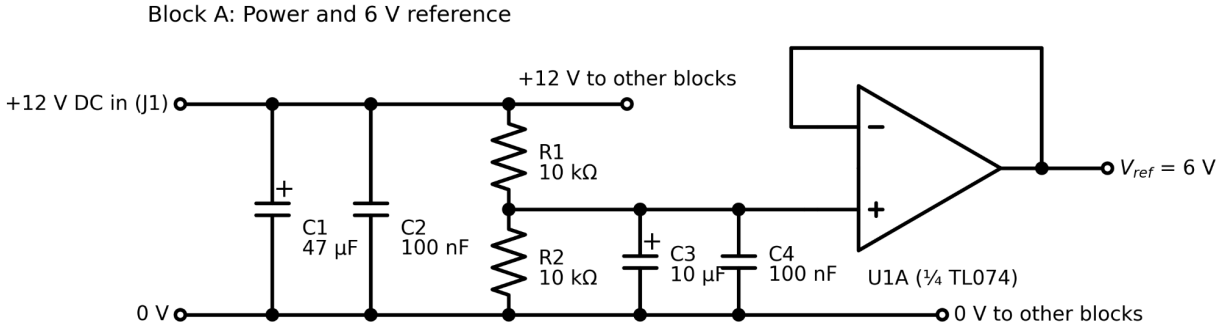


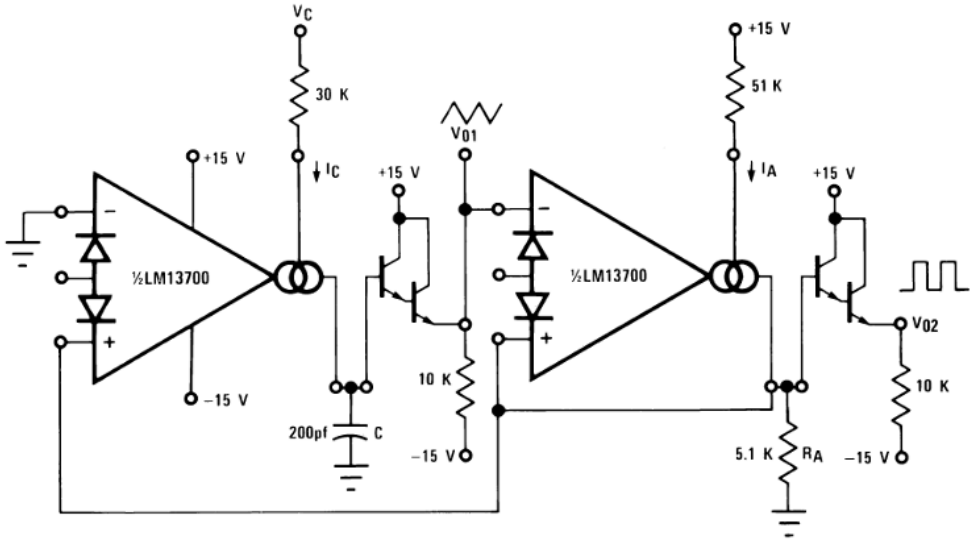
Figure 2. Power input, decoupling, and the buffered 6 V mid rail reference. U1 (TL074) supply pins: 4 to +12 V, 11 to 0 V.

Block B: OTA relaxation oscillator (square and triangle)

The oscillator follows the LM13700 datasheet circuit reproduced in Figure 3. The left OTA acts as a current source that charges and discharges the timing capacitor C with a constant current of $\pm I_C$, which produces a linear ramp, the triangle wave. The Darlington buffer inside the LM13700 passes this triangle to the right OTA, which is wired as a Schmitt trigger. Its output current $\pm I_A$ flows through R_A and sets switching thresholds of $\pm I_A * R_A$ around V_{ref}. When the ramp reaches a

threshold the trigger flips, the direction of I_C reverses, and the ramp changes slope. The trigger output is the square wave. Both waveforms therefore are always at the same frequency.

Block B: OTA relaxation oscillator, from the LM13700 datasheet (Texas Instruments)



$$f_{OSC} = \frac{I_C}{4CI_AR_A}$$

Figure 3. Triangle and square wave generator from the LM13700 datasheet (Texas Instruments). In our build, the ground and -15 V nodes become V_{ref} and 0 V respectively, and a +15 V becomes + 12 V.

The oscillation frequency is $f = I_C / (4\ C\ I_A\ R_A)$. We fix I_A and R_A so that the triangle amplitude I_AR_A is about 1 V peak. The LM13700 bias inputs sit about 1.4 V above the negative supply pin, so a resistor from the 12 V rail sets each bias current as I = (12 V - 1.4 V) / R. A 56 kΩ resistor to the bias input of the right OTA sets IA ≈ 0.19 mA, and RA is 5.1 kΩ. The frequency is then adjusted with IC alone: a 22 kΩ fixed resistor in series with a 1 MΩ linear potentiometer feeds the bias input of the left OTA, which gives IC from about 10 µA to 0.48 mA, a 46:1 sweep. The resulting first pass range is listed in Table 1.

Range (C)	**I_c = 10 µA**	**I_c = 0.48 mA**	**Intended use**
10 nF	≈ 270 Hz	≈ 12.4 kHz	Main audio range

Table 1. Predicted frequency range with I_AR_A ≈ 0.97 V. Values will be checked against simulation and measurement.

Because the amplitude of the triangle depends only on $I_A\ R_A$ and the frequency depends on I_C, changing the frequency does not change the amplitude at this stage. Fixing I_A and R_A is the key decision that decouples the two adjustments.

Block C: Triangle to sine shaper

The triangle wave is converted to a sine wave with the differential pair circuit from the Analog Devices activity, shown in Figure 4. The triangle is attenuated by R1 and R2 to a few hundred millivolts and applied to the base of Q1 while the base of Q2 sits at the reference. The collector current of a differential pair varies as $\tanh(V_{in} / 2\ V_T)$, where V_T is about 26 mV, so a triangle of the right amplitude drives the pair gently into its nonlinear region and the peaks are rounded into a shape that closely approximates a sine. The op-amp converts the two collector voltages to a single ended output, and the 100 pF feedback capacitor with the 4.7 kΩ feedback resistor forms a first order low-pass that filters the residual harmonics. The 500 Ω trimmer R6 balances the two halves of the pair and is adjusted for minimum even order distortion.

Block C: Triangle-to-sine shaper, from Analog Devices StudentZone, September 2021

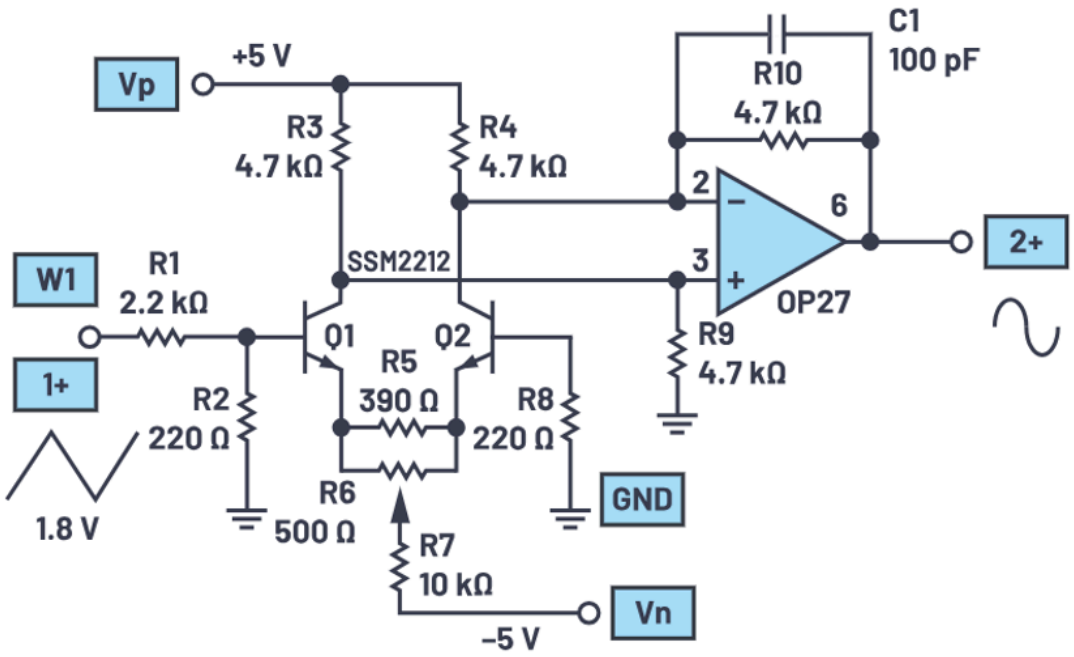


Figure 4. Differential pair triangle to sine converter (Analog Devices, StudentZone, September 2021).

To run this on our supply, the +5 V and -5 V connections become +12 V and 0V, the ground node at R2 and R9 becomes V_{ref}, and R7 returns to 0 V so that the tail current stays close to the original value. The reference expects a 3.6 V peak to peak triangle at the input. Our oscillator supplies about 2 V peak to peak, so R2 will be raised to about 390 Ω to restore the same base drive (the reference puts 3.6 V * 220/2420 ≈ 0.33 V peak to peak on the base. With 2 V peak to peak and R2 = 390 Ω the base sees 2 V * 390/2590 ≈ 0.30 V peak to peak), and the final value will be set by adjusting for the lowest third harmonic level on the FFT (on the oscilloscope if it

provides one, otherwise computed from exported waveform data). The reference uses an SSM2212 matched NPN pair, which we will use as well rather than hand matching discrete transistors. The OP27 in the reference will be replaced
with one section of the TL074.

Block D: Output amplifiers

Each of the three waveforms passes through its own inverting amplifier, shown in Figure 5, built from one section of the TL074. The non-inverting input is tied to V_{ref} so that the amplifier operates around the 6 V mid rail, and the gain is $-R_f / R_1$. With R_1 = 10 kΩ and R_f a 100 kΩ potentiometer in series with a 1 kΩ resistor, the gain is adjustable from roughly 0.1 to 10, which takes the 2 V peak to peak internal waveforms up to about 8 V peak to peak before the TL074 output stage begins to clip on the 12 V rail. The TL074 has built in output short circuit protection, so no series resistor is needed at the output. A DC blocking capacitor may be added at each output if the user needs a waveform centered on 0 V. We will decide this after the first integration test. Because the amplitude is set in this stage rather than in the oscillator, adjusting it does not disturb I_C or the timing capacitor, and the frequency stays fixed.

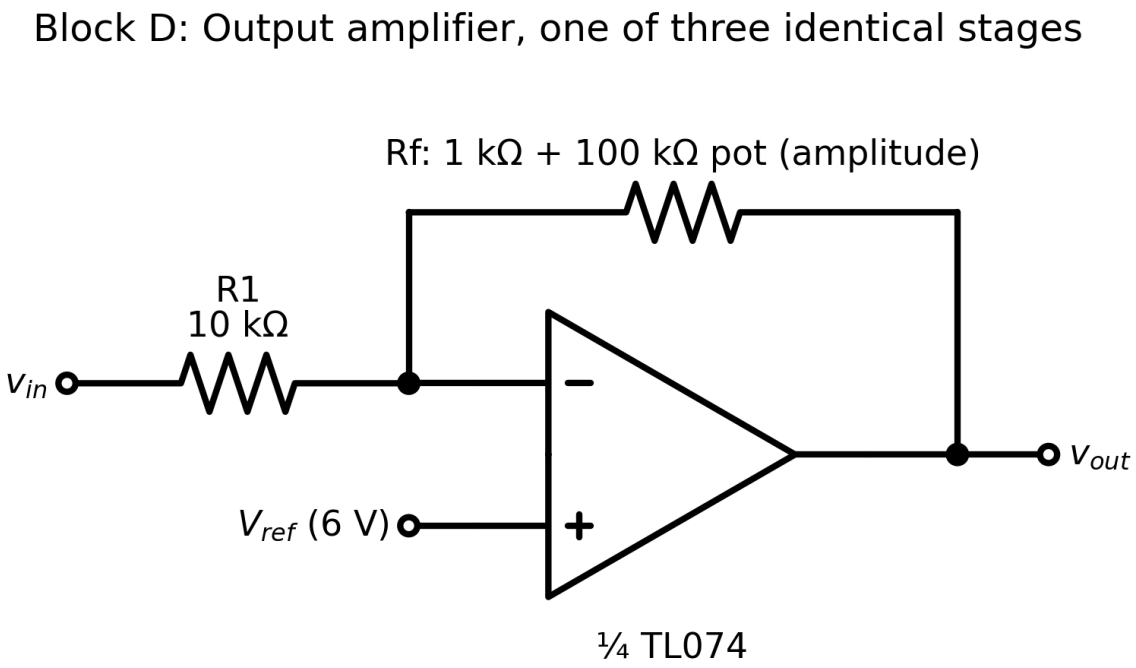


Figure 5. Inverting output amplifier referenced to V_{ref}. Sections A, B, and C of U2 (TL074) each drive one waveform. Supply pins as in Figure 2.

Milestones

We split the build into four tests, one per block, and wrote down what each one has to show before we move on. This way we have something to demonstrate at each checkpoint instead of waiting for the whole generator to work at the end.

Milestone 1 (week 6): Power and reference. Build Block A (Power) on the breadboard, then measure V_{ref} with the multimeter. Pass condition: V_{ref} within 6.0 V ± 0.1 V and changes by less than 50 mV when a 1 kΩ load is connected. In the same session, connect one LM13700 amplifier bias input through a resistor to the rail and confirm with the multimeter that the bias current matches (12 V - 1.4 V) / R within 20 percent. This tells us the single supply biasing of the OTA works before the oscillator is built.

Milestone 2 (week 8): Oscillator. Build Block B (Oscillator) and view the triangle and square outputs on the oscilloscope. Pass condition: both waveforms present across the full range of the frequency potentiometer, the triangle amplitude within 20 percent of the predicted 2 V peak to peak and roughly constant across the sweep, and the measured frequency within 20 percent of the formula at both ends of the range. We will record a table of frequency versus potentiometer position for the final report.

Milestone 3 (week 10): Sine shaper. Build Block C (Sine Shaper) and drive it from the lab function generator with a 1 kHz triangle, so that the shaper is tested independently of our oscillator. Adjust R6 for symmetry and R2 for amplitude while watching the FFT, on the oscilloscope if available or computed from exported waveform data. Pass condition: total harmonic distortion below 3 percent, which on the FFT means the largest harmonic is at least 30 dB below the fundamental (x dB = 20 log A, so 30 dB gives A = 0.032), and the shape remains acceptable from 300 Hz to 10 kHz. Then disconnect the bench generator, connect the triangle output from Milestone 2, and confirm the same result.

Milestone 4 (week 12): Full system. Add the three output amplifiers (Block D) and connect the four front panel potentiometers. Pass condition: all three waveforms available at the same time, and each amplitude adjustable from under 0.5 V to about 8 V peak to peak. The frequency should change by less than 1 percent when an amplitude potentiometer is swept from minimum to maximum. The waveforms should not change visibly when a 1 kΩ load is connected to any output.

Stretch goal (week 15): Transfer the circuit to PCB, mount the controls and a 12 V barrel jack in an enclosure, and repeat the Milestone 4 measurements on the finished unit.

Timeline

Week	Dates	Activity
4	9/21/26	Final proposal submitted. Confirm parts availability in the lab and place the order for the LM13700 and potentiometers.
5	9/28/26	Simulate the OTA oscillator and the sine shaper (LTspice or Falstad). Verify the frequency formula and choose the final resistor and capacitor values.
6	10/5/26	Build the power and V_{ref} block. Complete Milestone 1.
7	10/12/26	Build the LM13700 oscillator on the breadboard. Get the triangle and square waves running.
8	10/19/26	Add the frequency potentiometer. Complete Milestone 2 and record the frequency table.
9	10/26/26	Build the differential pair shaper. First distortion measurements with the bench generator.
10	11/2/26	Tune the shaper and connect it to the oscillator. Complete Milestone 3.
11	11/9/26	Build the three output amplifiers and wire the amplitude controls.
12	11/16/26	Integrate all blocks. Complete Milestone 4 and fix loading or noise problems.
13	11/23/26	Buffer week for debugging. Begin the PCB layout.
14	11/30/26	Assemble the final unit after Thanksgiving break. Repeat measurements and prepare the demo.
15	12/7/26	Final demo and report.

Materials

Expected to be available from the lab:

- Solderless breadboard and jumper wires
- 12 V bench supply, oscilloscope, and multimeter

- Resistors: 10 kΩ (8x), 4.7 kΩ (4x), 5.1 kΩ, 56 kΩ, 22 kΩ, 2.2 kΩ, 390 Ω, 220 Ω (2x), 1kΩ (3x)
- Capacitors: 100 nF (4x), 10 nF, 100 pF, 10 µF, 47 µF
- TL074 quad op-amp (or two TL072), two packages (five sections needed: V_{ref} buffer, shaper, and three output amplifiers)
- Small trim potentiometer, 500 Ω

To order:

Part	Qty.	Unit price	Source	Notes
LM13700M/NOPB dual OTA, SOIC-16	2	$1.70	Digikey LM13700M/NOPB-ND	One spare. TI no longer manufactures the DIP version, so the SOIC part is soldered to the adapter board below.
SOP-16 to DIP-16 adapter board	2	$0.05	Tayda A-7644	Carries the LM13700M on the breadboard. Needs two 8 pin header strips
1 MΩ linear potentiometer	1	$0.39	Tayda A-5616	Frequency control
100 kΩ linear potentiometer	3	$0.39	Tayda A-5519	Amplitude controls
12 V DC adapter and barrel jack	1	≈ $8	Amazon/Tayda	Final build only. Will use lab power supply during development
SSM2212RZ matched NPN pair, SOIC-8	1	$12.00	Mouser (584-SSM2212RZ)	Matched NPN pair for the shaper. Soldered to adapter board below.
SOP-8 to DIP-8 adapter board	1	$0.03	Tayda A-7643	Carries the SSM2212 on the breadboard. Needs two 4 pin header strips.

Table 2. Parts to order. Estimated total about $25. Prices marked ≈ are typical listings and will be confirmed during ordering.

Challenges

Our biggest unknown is the sine shaper. It depends on the two transistors being well matched and on the triangle amplitude landing close to the value the tanh curve requires. We will use the

SSM2212 matched pair, so the challenge in this block becomes careful calculation of the operating points and voltage swings around Vref, followed by fine tuning of R6 and R2 on the real circuit in the lab. Measuring the distortion is its own task, as the lab may not provide a usable FFT, so we plan to capture the waveform, export the data, and compute the FFT in MATLAB or LTspice, which means we also need to work on the data file conversion before Milestone 3. The second risk is single supply biasing. Every reference circuit we rely on is drawn for dual supplies, and moving the ground reference to V_{ref} changes where the OTA bias resistors and the shaper's tail resistor connect, so a mistake here produces no oscillation at all. Milestone 1 exists to catch this early, and the week 5 simulation will use the single supply topology rather than the original one. We are also unsure how the oscillator will behave at the ends of the frequency range, since the LM13700 bias current should stay below about 1 mA for linear operation and at very low bias currents leakage and the buffer's input current begin to affect the ramp. We have limited I_C to between 10 µA and 0.48 mA, but if the low end turns out to be unreliable we will raise C and claim a narrower range. We expect to ask the course staff for help with the FFT data export and processing, with choosing between perfboard and a PCB, and with confirming the final frequency range after simulation.